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Adjustable vehicular rearview mirror device

Abstract

An adjustable vehicular rearview mirror device for adjusting the rearview mirror to view blind spots of a vehicle includes a housing having a back wall and a perimeter wall being coupled to and extending forward from the back wall. A mirror is coupled to a front edge of the perimeter wall. The mirror includes a center section, a first lateral section, and a second lateral section. The first and second lateral sections each are pivotable about a corresponding vertical axis. A pair of levers includes a first lever and a second lever. The first lever is actuated to pivotally adjust the first lateral section relative to the corresponding vertical axis and the second lever is actuated to pivotally adjust the second lateral section relative to the corresponding vertical axis. A mount is coupled to an exterior surface of the back wall and mounts to an interior of a vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) Not Applicable
STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT
(2) Not Applicable
THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT
(3) Not Applicable
INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM
(4) Not Applicable
STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR
(5) Not Applicable
(g) BACKGROUND OF THE INVENTION
(1) Field of the Invention
(6) The disclosure relates to vehicle rearview mirrors and more particularly pertains to a new rearview mirror having adjustable sections of mirrors to facilitate the viewing of blindspots.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98
(7) The prior art relates to rearview mirrors and includes a variety of rearview mirrors being adjustable to enhance viewing a rear of a vehicle. Known prior art does not include a rearview mirror having a pair of sections being adjustable and a middle section being fixed such that the rearview mirror does not create a convex shape forming a fisheye view.
BRIEF SUMMARY OF THE INVENTION
(8) An embodiment of the disclosure meets the needs presented above by generally comprising a housing having a back wall being elongated and a perimeter wall being coupled to and extending forward from the back wall. The perimeter wall includes a top wall, a bottom wall, and a pair of side walls. A mirror is coupled to a front edge of the perimeter wall and is elongated. The mirror includes a center section, a first lateral section, and a second lateral section. The center section is fixed in the housing and the first and second lateral sections each are pivotable about a corresponding vertical axis. A pair of levers includes a first lever and a second lever. The first lever is coupled to the first lateral section and the second lever is coupled to the second lateral section of the mirror. The first lever is actuated to pivotally adjust the first lateral section relative to the corresponding vertical axis and the second lever is actuated to pivotally adjust the second lateral section relative to the corresponding vertical axis. A mount is coupled to an exterior surface of the back wall and is configured for mounting to an interior of a vehicle, wherein the mount mounts the housing to the interior of the vehicle. A display is coupled to the front edge of the perimeter wall and displays vehicular related indicia when actuated. An actuator is electrically coupled to the display and actuates the display when engaged by the user. A power supply is electrically coupled to the display and to the actuator.
(9) There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.
(10) The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)
(1) The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:
(2) FIG. **1** is a front isometric view of an adjustable vehicular rearview mirror device according to an embodiment of the disclosure.
(3) FIG. **2** is a rear isometric view of an embodiment of the disclosure.
(4) FIG. **3** is a front view of an embodiment of the disclosure.
(5) FIG. **4** is a rear view of an embodiment of the disclosure.
(6) FIG. **5** is a cross-sectional view of an embodiment of the disclosure taken along Line **5-5** of FIG. **4**.
(7) FIG. **6** is a partial cross-sectional view of an embodiment of the disclosure taken along Line **6-6** of FIG. **4**.
(8) FIG. **7** is an in-use view of an embodiment of the disclosure.
DETAILED DESCRIPTION OF THE INVENTION
(9) With reference now to the drawings, and in particular to FIGS. **1** through **7** thereof, a new

rearview mirror embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

(10) As best illustrated in FIGS. **1** through **7**, the adjustable vehicular rearview mirror device **10** generally comprises a housing **12** having a back wall **14** being elongated and a perimeter wall **16** being coupled to and extending forward from the back wall **14**. The perimeter wall **16** includes a top wall **18**, a bottom wall **20**, and a pair of side walls **22**. A mirror **24** is coupled to a front edge **26** of the perimeter wall **16** and is elongated. The mirror **24** includes a center section **28**, a first lateral section **30**, and a second lateral section **32**, wherein the center section **28** is positioned between the first **30** and second **32** later sections. As can be seen in the Figures, the mirror is comprised of forward side **34**, which is a forward facing mirrored side, and a rear side **36** that is comprised of a flexible material. The center section **28** is separated from the first **30** and second **32** sections and the flexible material allows the first **30** and second **32** sections to move relative to the center section **28**. This could also be achieved by hingedly coupling the first **30** and second **32** sections to the center section **28**. The center section **28** is fixed in the housing **12** relative to the first **30** and second **32** lateral sections. The first **30** and second **32** sections are generally pivotable, relative to the center section **28**, at vertical axes where the first **30** and second **32** sections abut the center section **28**.

(11) The adjustable vehicular rearview mirror device **10** includes a pair of levers **38** including a first lever **40** and a second lever **42**. The first lever **40** is coupled to the first lateral section **30** and the second lever **42** is coupled to the second lateral section **32** of the mirror **24**. The first lever **40** and the second lever **42** are actuated to adjust an angle in the forward sides of the first **30** and second **32** lateral sections relative to the center section **28**. An angle **43** formed in the forward sides of the first **30** and second **32** lateral sections with respect to the central section may be adjusted from 180° up to 225°, wherein the 180° denotes a flat mirror between formed with a corresponding one of the first **30** and second **32** lateral sections with the central section **28**. The center section **28** remains flat relative to the housing **12** such that the mirror **24** does not form a convex shape when the first **30** and second **32** lateral sections are adjusted so that the mirror **24** does not produce a fisheye view from forming the convex shape. However, it should be understood that entire mirror **24** may be pivotable within the housing **12** along a horizontal axis to utilize a dimmer feature **25** on the mirror **24** as is typical in conventional rearview mirror devices.

(12) A pair of tracks **44** is positioned within the housing **12** and is mounted to the top wall **18** of the housing **12**. A first track **46** of the pair of tracks **44** receives a top end of the first lever **40**, wherein the first lever **40** is slidable within the first track **46**. A second track **48** of the pair of tracks **44** receives an upper end of the second lever **42**, wherein the second lever **42** is slidable within the second track **48**. A pair of slots **50** is positioned on the bottom wall **20** of the housing **12** and extends therethrough. The pair of slots **50** includes a first slot **52** and a second slot **54**. A bottom end of the first lever **40** is slidably positioned within the first slot **52** and a lower end of the second lever **42** is slidably positioned within the second slot **54**. The user can actuate the first lever **40** by sliding the bottom end of the first lever **40** within the first slot **52** and the user can actuate the second lever **42** by sliding the lower end of the second lever **42** within the second slot **54**.

(13) A plurality of biasing members may be utilized within the housing **28** to bias the first **30** and second **32** sections back into a planar configuration with respect to the center section **28**. The biasing members in one embodiment may include multiple springs **56**. The plurality of springs **56** includes a first pair of springs **58** and a second pair of springs **60**. The first pair of springs **58** is coupled to the first lateral section **30** and the second pair of springs **60** is coupled to the second lateral section **32** of the mirror **24**. The first pair of springs **58** biases the first lateral section **30** to be positioned parallel with the center section **28** of the mirror **24** and the second pair of springs **60** biases the second lateral section **32** to be positioned parallel with the center section **28** of the mirror **24**. Each spring **56** of the plurality of springs **56** is positioned within the housing **12** and is coupled to an interior surface of the back wall **14**. The first pair of springs **58** is coupled to a back side of

the first lateral section **30** and the second pair of springs **60** is coupled to a back side of the second lateral section **32**.
(14) A mount **62** is coupled to an exterior surface of the back wall **14** and is configured for mounting to an interior **64** of a vehicle **66**, wherein the mount **62** mounts the housing **12** to the interior **64** of a vehicle **66**. The mount **62** typically mounts proximate to a top edge of a front windshield of the vehicle **66** typically either to the windshield itself or to a roof of the vehicle adjacent to the windshield. The mount **62** may include a pivot **68** to adjust the housing **12** relative to the interior of the vehicle when the housing **12** is mounted to the interior **64** of the vehicle **66**.
(15) A display **70** may be coupled to the front edge **26** of the perimeter wall **16** and displays vehicular related indicia **72** when actuated. The vehicular related indicia **72** may include time, temperature readings, cardinal directions of a compass, or any other vehicular related indicia **72** configured to provide information to the user. The display **70** typically comprises an electronic screen. An actuator **74** is electrically coupled to the display **70** and actuates the display **70** when engaged by the user. The actuator **74** comprises at least one switch and typically will include a plurality of push buttons. A power supply **76** is electrically coupled to the display **70** and to the actuator **74**. The power supply **76** may include at least one solar panel **78** mounted to the back wall **14**. The power supply **76** may also comprise a battery being positioned within the housing **12**. The battery may be rechargeable or replaceable. The power supply **76** may also be electrically coupled to a vehicle battery of the vehicle **66**, wherein the power supply **76** receives electricity from the vehicle battery.
(16) In use, the mount **62** mounts the housing **12** to the interior **64** of the vehicle **66**. The first lever **40** of the pair of levers **38** can be actuated to adjust the first lateral section **30** of the mirror **24** to enhance viewing a first-side of a rear of the vehicle **66**. The second lever **42** of the pair of levers **38** can be actuated to adjust the second lateral section **32** of the mirror **24** to enhance viewing a second-side of the rear of the vehicle **66**. The center section **28** of the mirror **24** is fixed to view a center of the rear of the vehicle **66**. The user may engage with the actuator **74** to actuate the display **70** to display vehicular related indicia **72**.
(17) With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.
(18) Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. An adjustable rearview mirror device comprising: a housing having a back wall being elongated and a perimeter wall being coupled to and extending forward from said back wall, said perimeter wall including a top wall, a bottom wall, and a pair of side walls; a mirror being coupled to a front edge of said perimeter wall and being elongated, said mirror including a center section, a first lateral section, and a second lateral section, said center section being fixed in said housing, said

first and second lateral sections each being pivotable relative to said center section; a pair of levers including a first lever and a second lever, said first lever being coupled to said first lateral section of said mirror and said second lever being coupled to said second lateral section of said mirror, said first lever being actuated to pivotally adjust said first lateral section relative to said center section and said second lever being actuated to pivotally adjust said second lateral section relative to said center section; a mount being coupled to an exterior surface of said back wall, said mount being configured for mounting to an interior of a vehicle, wherein said mount mounts said housing to the interior of the vehicle; a display being coupled to said front edge of said perimeter wall and displaying vehicular related indicia when actuated; an actuator being electrically coupled to said display and actuating said display when engaged by a user; and a power supply being electrically coupled to said display and to said actuator.

2. The adjustable rearview mirror device of claim 1, further including a pair of tracks being positioned within said housing and being mounted to said top wall of said housing, a first track of said pair of tracks receiving a top end of said first lever, wherein said first lever being slidably positionable within said first track, a second track of said pair of tracks receiving a top end of said second lever, wherein said second lever being slidably positionable within said second track.

3. The adjustable rearview mirror device of claim 2, further including a plurality of biasing members to bias said first and second sections into a planar configuration with respect to said center section, said biasing members comprising a plurality of springs, each spring of said plurality of springs being positioned within said housing and being coupled to an interior surface of said back wall, said plurality of springs including a first pair of springs and a second pair of springs, said first pair of springs being coupled to said first lateral section and said second pair of springs being coupled to said second lateral section of said mirror, said first pair of springs biasing said first lateral section to be positioned parallel with said center section of said mirror and said second pair of springs biasing said second lateral section to be positioned parallel with said center section of said mirror, said first pair of springs being coupled to said back side of said first lateral section and said second pair of springs being coupled to said back side of said second lateral section.

4. The adjustable rearview mirror device of claim 1, wherein said display comprises an electronic screen.

5. The adjustable rearview mirror device of claim 4, wherein said actuator comprises at least one switch.

6. The adjustable rearview mirror device of claim 1, wherein said power supply includes at least one solar panel, at least one solar panel being mounted to said back wall.

7. An adjustable rearview mirror device comprising: a housing having a back wall being elongated and a perimeter wall being coupled to and extending forward from said back wall, said perimeter wall including a top wall, a bottom wall, and a pair of side walls; a mirror being coupled to a front edge of said perimeter wall and being elongated, said mirror including a center section, a first lateral section, and a second lateral section, said center section being positioned between said first and second lateral sections, said center section being fixed in said housing, said first and second lateral sections each being pivotable relative to said center section; a pair of levers including a first lever and a second lever, said first lever being coupled to said first lateral section of said mirror and said second lever being coupled to said second lateral section of said mirror, said first lever being actuated to pivotally adjust said first lateral section relative to said center section and said second lever being actuated to pivotally adjust said second lateral section relative to said center section; a pair of tracks being positioned within said housing and being mounted to said top wall of said housing, a first track of said pair of tracks receiving a top end of said first lever, wherein said first lever being slidably positionable within said first track, a second track of said pair of tracks receiving a top end of said second lever, wherein said second lever being slidably positionable within said second track; a pair of slots being positioned on said bottom wall of said housing and extending therethrough, said pair of slots including a first slot and a second slot, a bottom end of

said first lever being slidably positioned within said first slot and a bottom end of said second lever being slidably positioned within said second slot; a plurality of biasing members to bias said first and second sections into a planar configuration with respect to said center section, said biasing members comprising a plurality of springs, each spring of said plurality of springs being positioned within said housing and being coupled to an interior surface of said back wall, said plurality of springs including a first pair of springs and a second pair of springs, said first pair of springs being coupled to said first lateral section and said second pair of springs being coupled to said second lateral section of said mirror, said first pair of springs biasing said first lateral section to be positioned parallel with said center section of said mirror and said second pair of springs biasing said second lateral section to be positioned parallel with said center section of said mirror, said first pair of springs being coupled to said back side of said first lateral section and said second pair of springs being coupled to said back side of said second lateral section; a mount being coupled to an exterior surface of said back wall, said mount being configured for mounting to an interior of a vehicle, wherein said mount mounts said housing to the interior of the vehicle; a display being coupled to said front edge of said perimeter wall and displaying vehicular related indicia when actuated, said display comprising an electronic screen; an actuator being electrically coupled to said display and actuating said display when engaged by a user, said actuator comprising at least one switch; and a power supply being electrically coupled to said display and to said actuator, said power supply including at least one solar panel, at least one solar panel being mounted to said back wall.